Data-Driven Productivity Optimization

Analyzing 6 Weeks of Performance Metrics

Project Objective

Goal: Optimize work schedule based on empirical performance data

Challenge:

- Multiple concurrent priorities across different domains
- Academic coursework (Python, SQL, Statistics, Power BI)
- Content development (lesson plans, curricula)
- Professional development tasks
- Personal projects

Question: How can schedule design maximize output quality

while managing capacity effectively?

Methodology

Data Collection & Analysis Framework

Data Collection Method

Tracking Period: October 31 - November 26, 2025 (6 weeks)

Data Sources:

- AI-assisted activity logging system
- Timestamped work session records
- Output volume measurement per session

Variables Measured:

- Duration (hours per session)
- Output quality (1-10 scale, post-session evaluation)
- Activity category and complexity level
- Time of day (morning/afternoon/evening)
- Tasks initiated and completed

Performance Metrics: Calculation Methods

Energy (Objective)

```
Energy = (Output Quality × Task Complexity) ÷ 10
```

Productivity Score

```
Productivity = (Hours × Energy × Quality) ÷ 100
```

Completion Rate

```
Completion = (Tasks Completed ÷ Tasks Started) × 100
```

Task Efficiency

```
Efficiency = Tasks Completed ÷ Hours Worked
```

Task Complexity Scoring System

Complexity scores assigned by cognitive demand:

High Complexity (10/10):

- Python programming, Database development
- Complex analytical work requiring deep focus

Medium-High Complexity (8-9/10):

- Statistics, Power BI, Privacy analysis
- Structured problem-solving tasks

Medium Complexity (6-7/10):

- ESL content creation, Business development
- Creative work with moderate cognitive load

Low Complexity (2-5/10):

Dataset Summary: 6-Week Period

94 hrs

Total Hours

57

Work Sessions

78 of 91

Task Completion

5.9/10

Avg Energy

8.3/10

Avg Quality

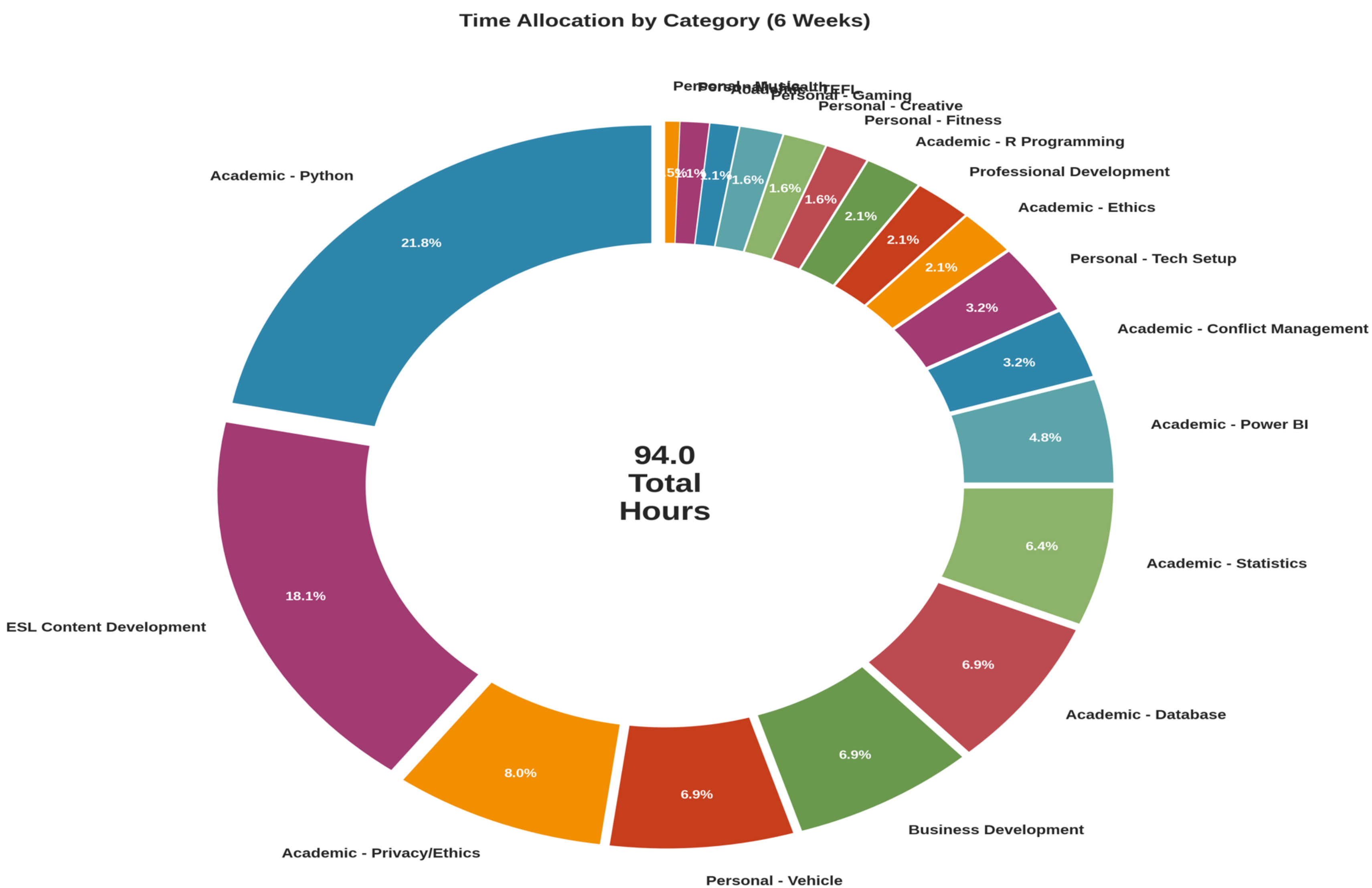
85.7%

Completion Rate

Findings

Data Analysis Results

Finding 1: Time Distribution Across Categories



Finding 2: Two Types of Productive Capacity

Morning (8am-12pm):

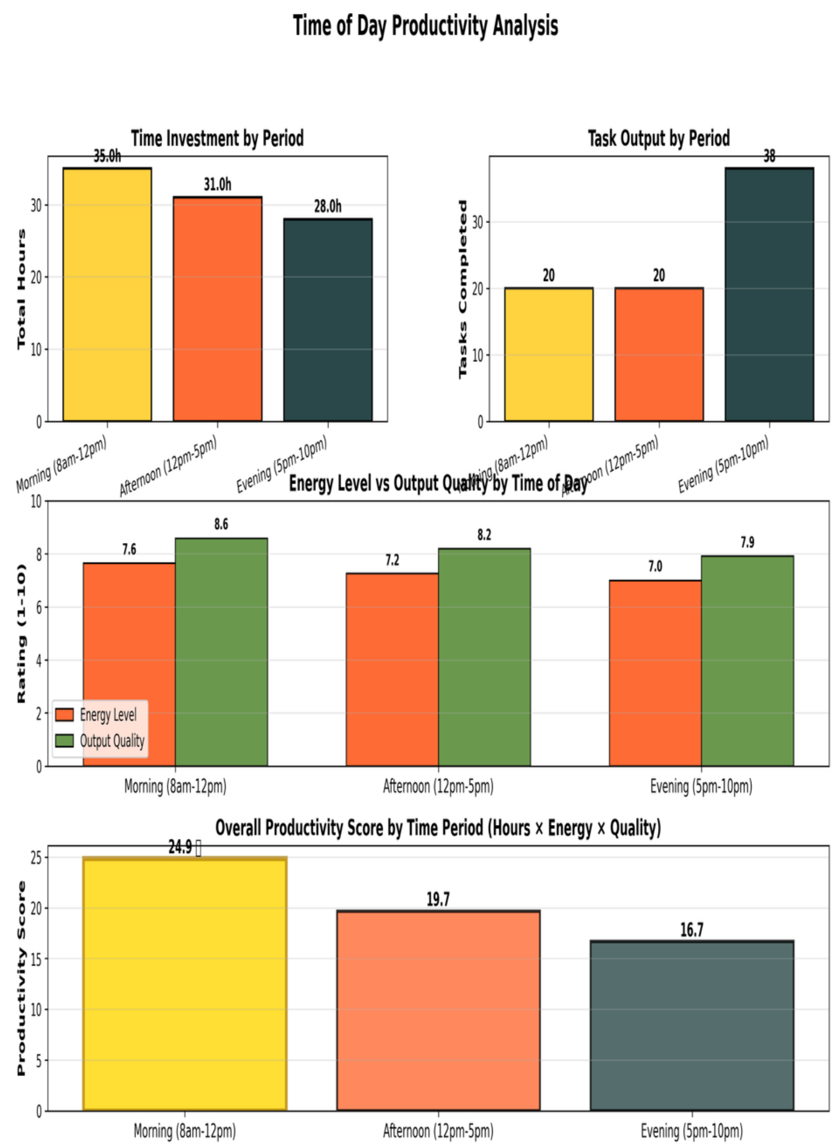
- Energy: 7.9/10 (Focus Capacity)
- Quality: 8.6/10
- Tasks: 20 (Output Capacity: 3.4/10)
- Avg Complexity: 9.3/10

Afternoon (12pm-5pm):

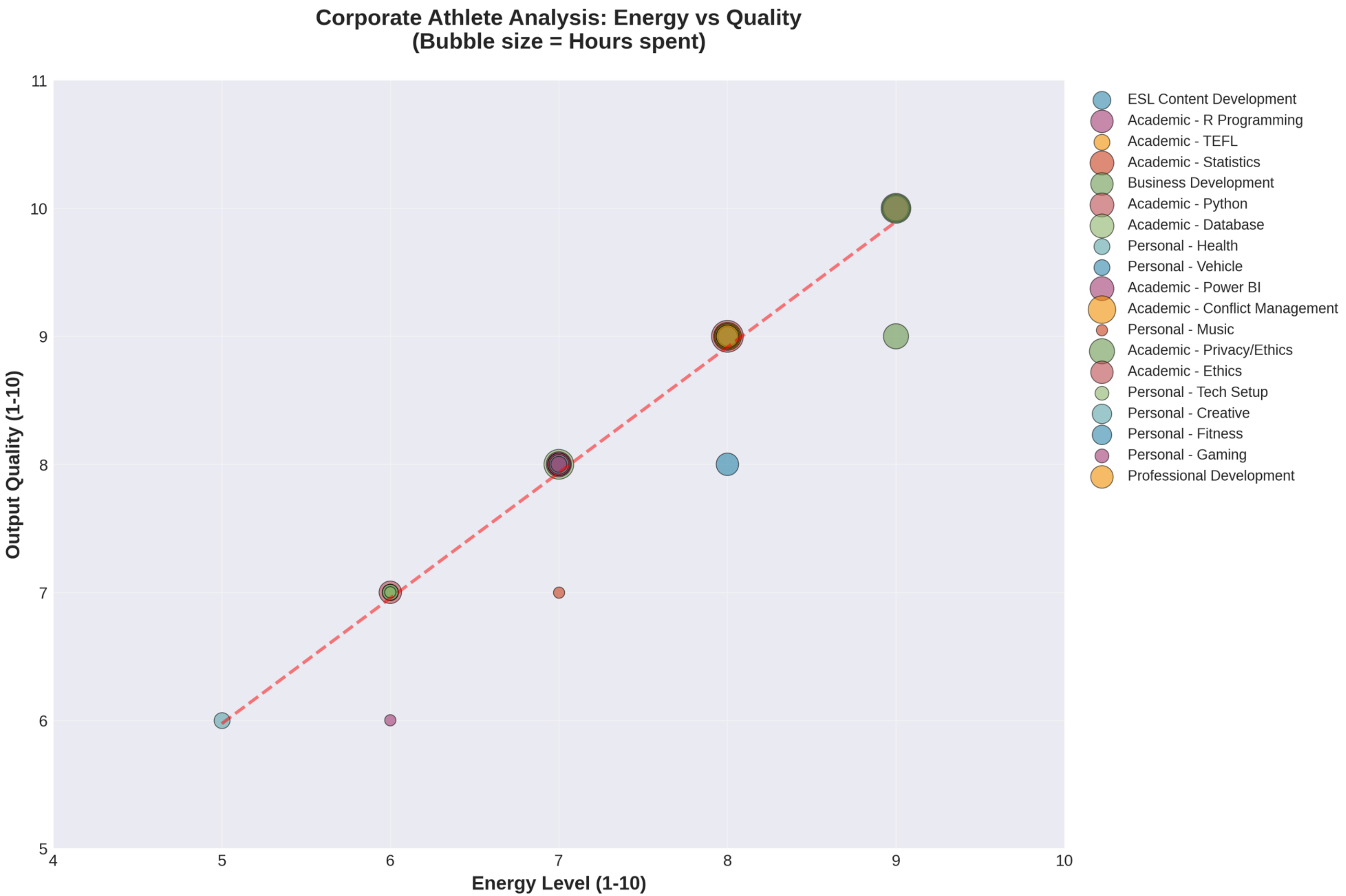
- Energy: 6.3/10 (Focus Capacity)
- Quality: 8.2/10
- Tasks: 20 (Output Capacity: 3.6/10)
- Avg Complexity: 7.7/10

Evening (5pm-10pm):

- Energy: 3.7/10 (Focus Capacity)
- Quality: 7.9/10
- Tasks: 38 (Output Capacity: 7.5/10)
- Avg Complexity: 4.8/10



Finding 3: Energy Reflects Cognitive Deployment



Finding 4: Task Completion Performance

Task Flow Analysis:

• 91 tasks initiated

• 78 tasks completed

• 85.7% completion rate

• 13 tasks in progress (14.3%)

Work-In-Progress Metrics:

• Average daily WIP: 0.6 tasks

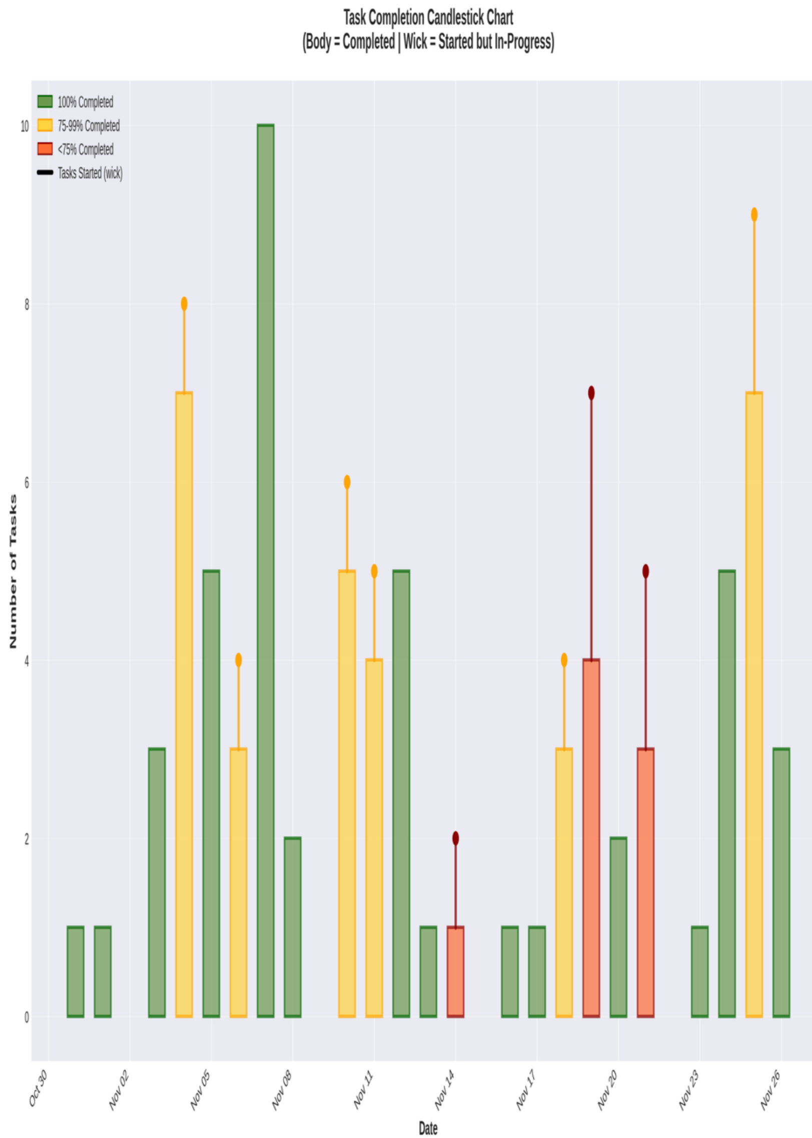
• Maintained low WIP throughout period

Completion Pattern:

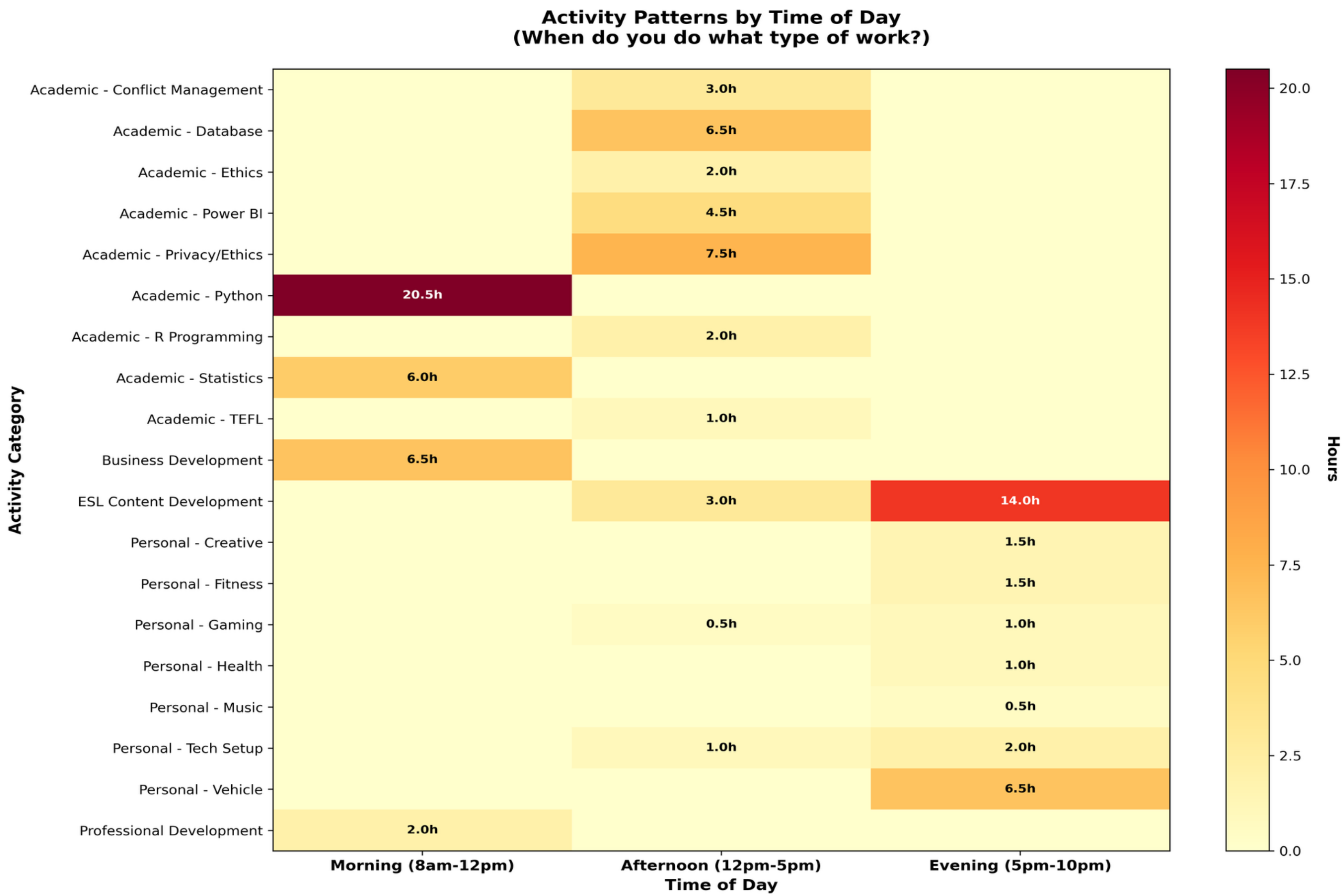
• Consistent daily closure rate

• Most sessions: same-day completion

• Complex projects: 2-3 day completion



Finding 5: Work Type Naturally Aligns with Time Period



Analysis

Interpreting Performance Patterns

Pattern Analysis: Focus vs. Volume Capacity

1. Morning Focus Advantage
   - Energy (focus capacity): 7.9/10 - highest cognitive capacity
   - Handles complex work: avg complexity 9.3/10
   - Productivity for complex work: 124% higher than evening
   - Optimal for: Python, Database, Strategic Analysis

2. Evening Volume Advantage
   - Output capacity: 7.5/10 - highest task throughput
   - Completed 38 tasks vs 20 in morning
   - Lower complexity work: avg 4.8/10
   - Optimal for: Content creation, Admin, Quick tasks

3. Task Rotation & Rest Strategy
   - Alternating work types prevents fatigue
   - Strategic 20-minute rest intervals utilized

Productivity Metrics: Complex Work Performance

0.731/hr

Morning Productivity

0.592/hr

Afternoon Productivity

0.326/hr

Evening Productivity

+124%

Morning Advantage

Improvement Opportunity: Optimize Time Allocation

Current Morning Allocation:

• 35 hours over 6 weeks (37.2% of total)

• Highest focus capacity window underutilized for complex tasks

Observed Misalignment:

• Complex analytical work sometimes scheduled in evening

• High-value tasks occurring during lower-capacity periods

• Simple tasks occupying prime morning hours

Optimization Potential:

• Increase morning allocation for complex work: 35 → 45+ hours

• Reserve peak cognitive capacity for priority tasks

• Move high-volume simple tasks to evening

Expected Impact:

Optimization

Data-Based Schedule Design

Proposed Schedule: Capacity-Task Alignment

Morning Block (8am-12pm) - Focus Capacity Zone

• Energy: 7.9/10 (highest cognitive capacity)

• Schedule: Complex analytical work

- Python programming, Database development

- Statistical analysis, Strategic planning

• Rationale: Quality (8.6) × Complexity (9.3) = Peak performance

Afternoon Block (12pm-5pm) - Implementation Zone

• Energy: 6.3/10 (moderate focus capacity)

• Schedule: Structured technical tasks

- Power BI development, SQL queries

- Hands-on project work

• Rationale: Execute plans made in morning peak hours

Evening Block (5pm-10pm) - Volume Capacity Zone

Implementation: Phased Schedule Transition

Phase 1: Immediate Reallocation (Week 1-2)

• Block 8am-12pm for complex analytical work only

• Move routine tasks to afternoon/evening slots

• Track energy and quality metrics daily

Phase 2: Refinement & Adjustment (Week 3-4)

• Monitor productivity scores using same formulas

• Adjust time block boundaries based on performance data

• Implement 20-min rest intervals as needed

Phase 3: Validation & Measurement (Week 5-6)

• Compare before/after productivity scores

• Calculate actual improvement percentage

• Document quality improvements on complex deliverables

Projected Performance Improvements

Quantified Targets:

1. Productivity Score Increase
   - Current weighted average: 0.548/hr
   - Target: 0.620+/hr (13.1% improvement)
   - Method: Shift 15+ hours of complex work to morning

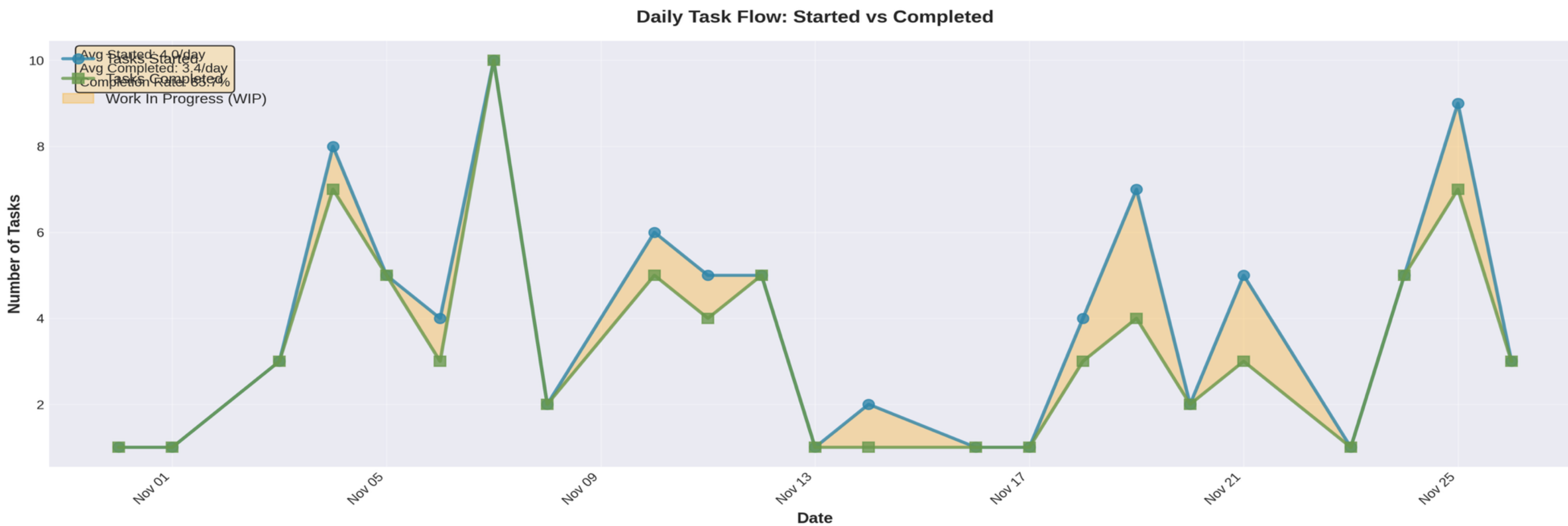
2. Quality Output on Critical Tasks
   - Leverage 124% morning advantage for priority work
   - Maintain 8.6/10 quality on analytical deliverables
   - Reduce quality variance across time periods

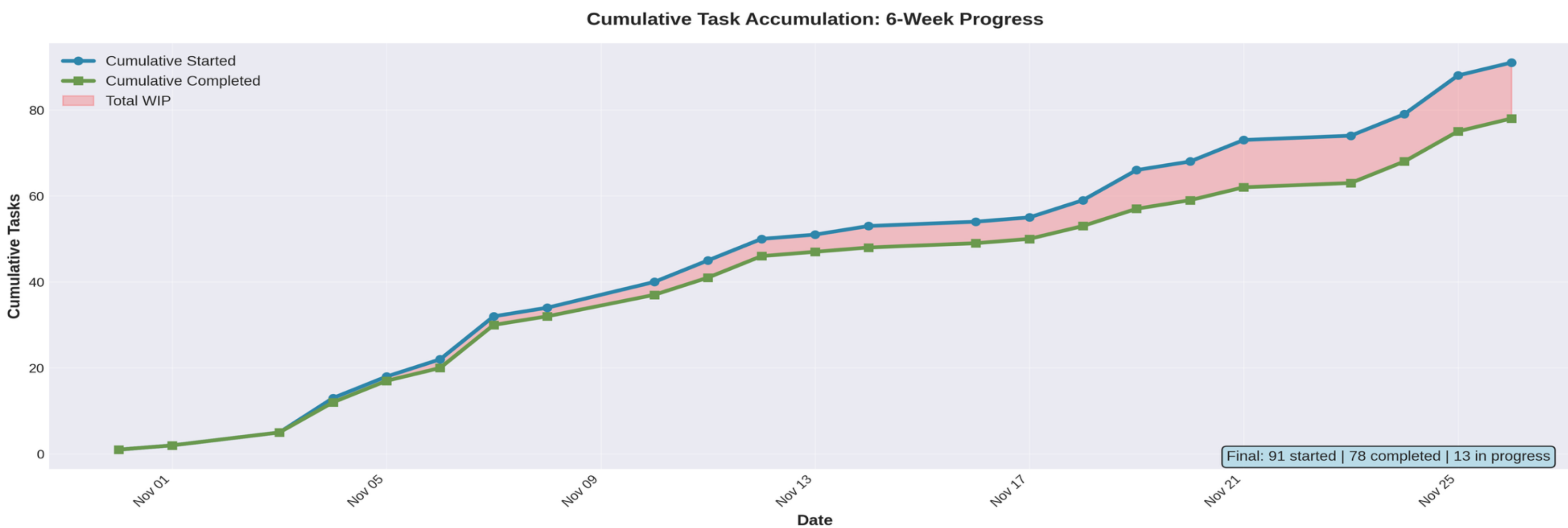
3. Task Management Efficiency
   - Sustain 85%+ completion rate
   - Keep WIP below 1.0 task/day average

Performance Data

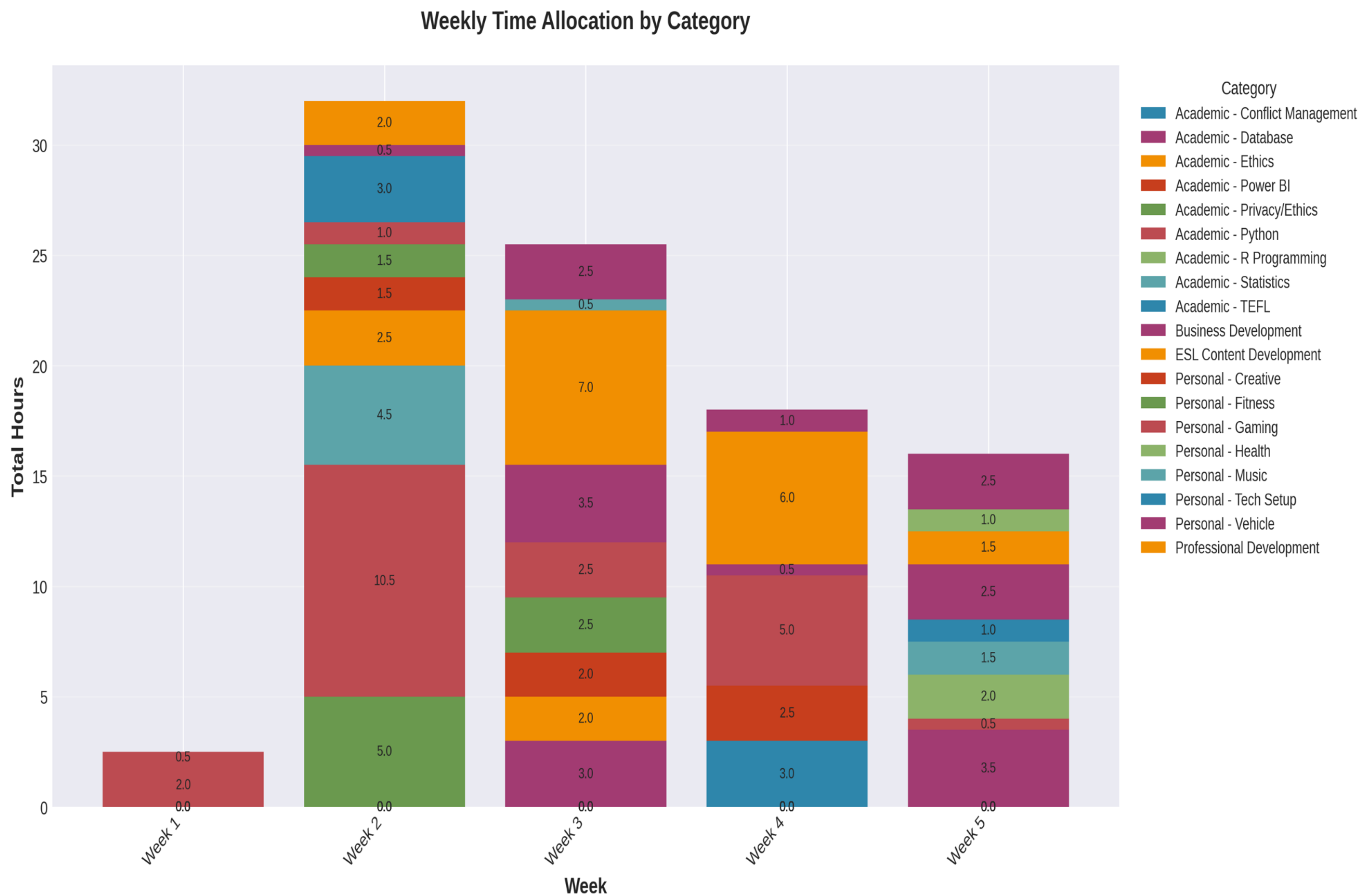
6-Week Documented Results

Task Completion: Sustained Performance Pattern





Weekly Performance Distribution



Measured Outcomes: Baseline Performance

Performance Metrics (6-Week Period):

✓ 94 hours of documented productive work

✓ 57 work sessions with measured energy levels

✓ 78 tasks completed (85.7% completion rate)

✓ 8.3/10 average output quality sustained

✓ 5.9/10 average energy (objective measurement)

✓ 0.6 tasks WIP per day (focused execution)

Skill Development (Hours → Proficiency):

- Python/Data Analysis: 20.5 hrs → 8.7/10 quality
- Database Management: 6.5 hrs → 8.0/10 quality
- Business Intelligence: 4.5 hrs → 8.0/10 quality
- Data Privacy/Ethics: 7.5 hrs → 9.3/10 quality

Capacity Distribution:

Conclusions & Recommendations

Data-Supported Findings:

1. Cognitive capacity varies significantly by time of day
   - Morning: 7.9/10 focus capacity (124% productivity advantage)
   - Evening: 7.5/10 volume capacity (best for high-throughput work)
   - Recommendation: Match task complexity to capacity profile

2. Objective energy measurement validates schedule design
   - Energy = (Quality × Complexity) ÷ 10
   - Captures cognitive deployment, not subjective feeling
   - Recommendation: Use formula to evaluate schedule effectiveness

3. Task completion discipline improves overall performance
   - Low WIP (0.6 tasks/day) correlates with 85.7% completion
   - Recommendation: Limit concurrent tasks, focus on closure